

VISHNU SANJAY KUMAR P

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Career Objective

Artificial Intelligence and Data Science undergraduate with hands-on experience in Machine Learning, IoT systems, and full-stack development using MERN and Flutter. Built real-world AI solutions across predictive maintenance, healthcare wearables, and agricultural disease detection systems. Seeking Software Developer or Machine Learning Engineer roles to develop scalable intelligent systems.

Academic Details

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| • Francis Xavier Engineering College, Tirunelveli
B.Tech – Artificial Intelligence and Data Science
CGPA: 8.4/10 | 2023–2027 |
| • Muthamil Public School(CBSE), Tirunelveli, Tamil Nadu (HSC)
Percentage: 64% | 2023 |
| • Muthamil Public School(CBSE), Tirunelveli, Tamil Nadu (SSLC)
Percentage: 65% | 2021 |

Technical Skills

- Programming Languages: Java, Python, JavaScript, SQL.
- Frame Work: Spring Boot, Flutter ,React.js ,Node.js, Express.js, MERN Stack.
- Machine Learning & AI: Machine Learning, Deep Learning, TensorFlow, Scikit-Learn, CNN, ANN.
- Databases : MySQL, MongoDB.
- Tools: Git, GitHub, MQTT, Grafana, Power BI, Android Studio, Jupyter Notebook.

Project Development

Hardware Projects:

MachinaGuard – AI-Powered Predictive Maintenance System

Python, **Flask**, Scikit-Learn, MySQL, IoT Sensors

- Built an AI-based predictive maintenance system using industrial IoT sensor data for equipment health monitoring.
- Developed ML classification models to predict machine failure and reduce downtime.

MobiSafe – AI-Powered Fall Detection and Emergency Alert System

ESP32, TinyML, Edge Impulse, Neural Networks, GPS

- Developed a TinyML-based wearable system for real-time fall detection using IMU sensor data.
- Deployed optimized neural network model on ESP32 for edge AI inference.
- Integrated GPS-based emergency alert system for instant notifications.

OnioVault – Smart Modular IoT-Based Onion Cold Storage System

ESP32, MQTT, IoT Sensors

- Designed an IoT-based cold storage system for onion preservation using real-time environmental monitoring.

Software Projects:

Memory Aid – AI-Powered Cognitive Assistance Mobile Application

Flutter, Python, MySQL, Machine Learning, **Ollama**

- Built a Flutter-based AI mobile app for task scheduling, reminders, and cognitive assistance.
- Implemented real-time data sync with backend database for personalized notifications.
- Designed AI-driven reminder system to improve user productivity and memory support.

WheatGuard – AI-Powered Wheat Disease Detection System [Live](#)

CNN, **TensorFlow**, Computer Vision

- Developed CNN-based deep learning model for wheat disease classification using leaf image dataset.
- Trained TensorFlow model to detect multiple disease categories with high accuracy.
- Built image preprocessing pipeline for improved model performance.

ZenPath – Stress Prediction System [Live](#)

FastAPI, Random Forest, PostgreSQL, Machine Learning, REST API

- Built a FastAPI backend with REST APIs for user authentication, stress prediction, and history retrieval.
- Integrated a Random Forest model to predict stress levels and stored prediction history in PostgreSQL for future analysis.

Internship

Srishti Innovative Pvt. Ltd. – Artificial Intelligence & IoT Intern Dec 2025 – Jan 2026

- Developed an AI-based anomaly detection system for real-time traffic and intrusion prediction using machine learning models with dashboard-based monitoring and alerts.
- Implemented a computer vision module using ArUco marker detection for object tracking, integrated with IoT.

Marcello Tech (AICTE) – Machine Learning, AI & Deep Learning Intern Jul 2024 – Sep 2024

- Built and trained machine learning and deep learning models using Python for real-world datasets.
- Handled full ML lifecycle including data preprocessing, model training, testing, and performance improvement.

Quantanics TechServ Pvt. Ltd. – Embedded Systems & IoT Intern Apr 2024 – May 2024

- Worked on embedded IoT systems using NodeMCU and Raspberry Pi for sensor data acquisition and real-time monitoring applications.
- Developed IoT data pipelines for machine learning training and explored deployment concepts on edge devices.

Certifications

Artificial Intelligence / Machine Learning

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| • Machine Learning – Stanford Online (Coursera) | Feb 2026 |
| • Machine Learning – Pantech Solutions | Jun 2024 |
| • Large Language Models (LLM) – NPTEL, 55% | Jan–Apr 2025 |

Software Development / Full Stack

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| • MERN Stack Development – Packt (Coursera) | May 2026 |
| • Spring Framework for Java Development – Coursera SkillUp | Apr 2026 |
| • Java Development with Databases – Coursera SkillUp | Apr 2026 |
| • Cloud Native, Microservices, Containers, DevOps & Agile – IBM (Coursera) | Apr 2026 |

IoT / Embedded Systems

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| • Embedded Systems & IoT – Offline Certification | Apr–May 2024 |
| • Industry 4.0 & Industrial IoT – NPTEL,(Elite, 72) | Jan-Apr 2025 |

Programming / CS Fundamentals

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| • The Joy of Computing Using Python – NPTEL, 2025 (Elite, 65%) | Jan-Apr 2025 |
| • Privacy and Security in Online Social Media – NPTEL(Elite,67%) | Jul-Oct 2025 |

Achievements & Competitions

International Recognition

- **Global Nominee** – NASA Space Apps Challenge 2025, selected among 11,500+ global
- **Honorable Mention** – Yesist’12 International Innovation Contest 2024, Tunisia, North Africa.

National-Level Competitions

- **Winner** – Hack the Mountain 2024 (National Online Hackathon), recognized for AI-based problem-solving and innovative solution development.
- **Winner** – VISAI 2025, AI-based Predictive Maintenance System competition.

Institutional-Level Achievements

- **Winner** – HACK2HIRE AI Challenge 2025 (College-Level Hackathon), developed an AI-powered Memory Aid mobile application for cognitive assistance.

Innovation & Funding

- **Innovation Grant Recipient** – ₹2.5 Lakhs, EDII Tamil Nadu – Innovation Voucher Programme (IVP) for AI/IoT-based innovation project development.

Languages Known

Tamil, English